Javad Komijani

Curriculum Vitae

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Professional Summary

Computational physicist and machine-learning researcher specializing in lattice gauge theory and scientific machine learning. Experienced in developing GPU-accelerated machine learning frameworks and differentiable computing tools using PyTorch. Strong background in mathematical physics, effective field theories, statistical modeling, and Monte Carlo methods.
Research focus: symmetry-aware (equivariant) machine learning and scientific computing for physical systems, including high-energy physics and cosmology.

Education

2015 **PhD in Physics**, *Washington University in St. Louis*
Thesis: *Topics in Lattice Gauge Theory and Theoretical Physics*
- Developed statistical models based on effective field theories.
- Built Bayesian inference pipelines for extracting physical parameters from large-scale lattice-QCD simulations using the developed theoretical models.
- Introduced a novel definition for eigenvalues in the context of nonlinear problems.

2009 **MSc in Electrical Engineering**, *University of Tehran*
- Focus: electromagnetic waves and fields, signal processing, statistical analysis.

2006 **BSc in Electrical Engineering**, *University of Tehran*

Research and Teaching Experience

2021–Present **Postdoctoral Researcher**, *ETH Zurich*
- Developed a novel score-matching framework for diffusion models on non-Abelian Lie groups for lattice gauge theory, enabling generative modeling of SU(N) gauge configurations.
- Implemented GPU-accelerated distributed training infrastructure for large-scale lattice simulations.
- Built installable PyTorch extensions for differentiable linear algebra and adjoint-based ODE solvers supporting Lie-group-valued systems.
- Contributed machine-learning modules to the EU interTwin Digital Twin Engine, integrating normalizing flows for lattice QCD simulations.
- Supervised PhD and Master's students in scientific machine learning and computational physics.
- Collaborated on international projects in scientific machine learning and high-energy physics.
- Performed teaching activities, including teaching quantum mechanics.

2019–2020 **Postdoctoral Researcher**, *University of Tehran*
- Applied lattice-QCD simulations and Bayesian analysis to determine Standard Model parameters.
- Organized a workshop on lattice field theory, teaching lattice methods to students from various universities in Tehran.

2017–2018 **Postdoctoral Researcher**, *University of Glasgow*
- Focused on determination of various form factors and quark condensates, using lattice-QCD Monte-Carlo simulations and Bayesian analysis.
- Collaborated in teaching activities, including scientific computing.

2015–2017 **Postdoctoral Researcher**, *Technical University of Munich*

- Focused on determination of quark masses and various decay constants, using lattice-QCD Monte-Carlo simulations and Bayesian analysis.
- Collaborated in teaching activities, including teaching quantum field theory and elementary particle physics.

Scientific Software and ML Infrastructure

lattice_ml **PyTorch scientific software package**

- Diffusion-based generative modeling on Lie groups and gauge theories.
- Adjoint-based differentiable ODE solvers for scalar and Lie-group-valued systems.
- Differentiable SVD and eigendecomposition of unitary matrices.

normflow PyTorch software package for normalizing flows on Lie groups and lattice gauge theory.

interTwin Contributed normalizing-flow-based ML modules to the EU interTwin Digital Twin Engine for lattice QCD simulations.

Selected Projects

SU(N) Diffusion Extended score-matching diffusion models to Lie groups for lattice gauge theory and applied generative modeling of SU(3) gauge configurations in two and four dimensions.

Quark Masses Proposed and implemented a method to calculate heavy quark masses; led to precise determination of the up-, down-, strange-, charm-, and bottom-quark masses using the MILC highly improved staggered-quark ensembles.

MRS Mass Defined a novel quark mass scheme, the minimal-renormalon-subtracted (MRS) mass, which is free from the shortcomings of other quark masses and provides a more robust foundation for precise quark mass determination.

Eigenvalues Generalized eigenvalue problems to non-linear cases using properties of solutions to the Schrödinger equation.

Technical Skills

Programming	Python, C++, Cython, Bash
ML/AI	PyTorch, Diffusion Models, Normalizing Flows, Score Matching, Distributed Training
Mathematics	Differential Equations, Linear Algebra, Optimization, Statistical Modeling
Tools	Linux, Git, SQL, Pandas
Scientific	NumPy, SciPy, Scikit-learn, Monte Carlo Methods

Language Skills

English	Fluent
German	Basic (A2)

Publications

Full list of peer-reviewed publications: available here.